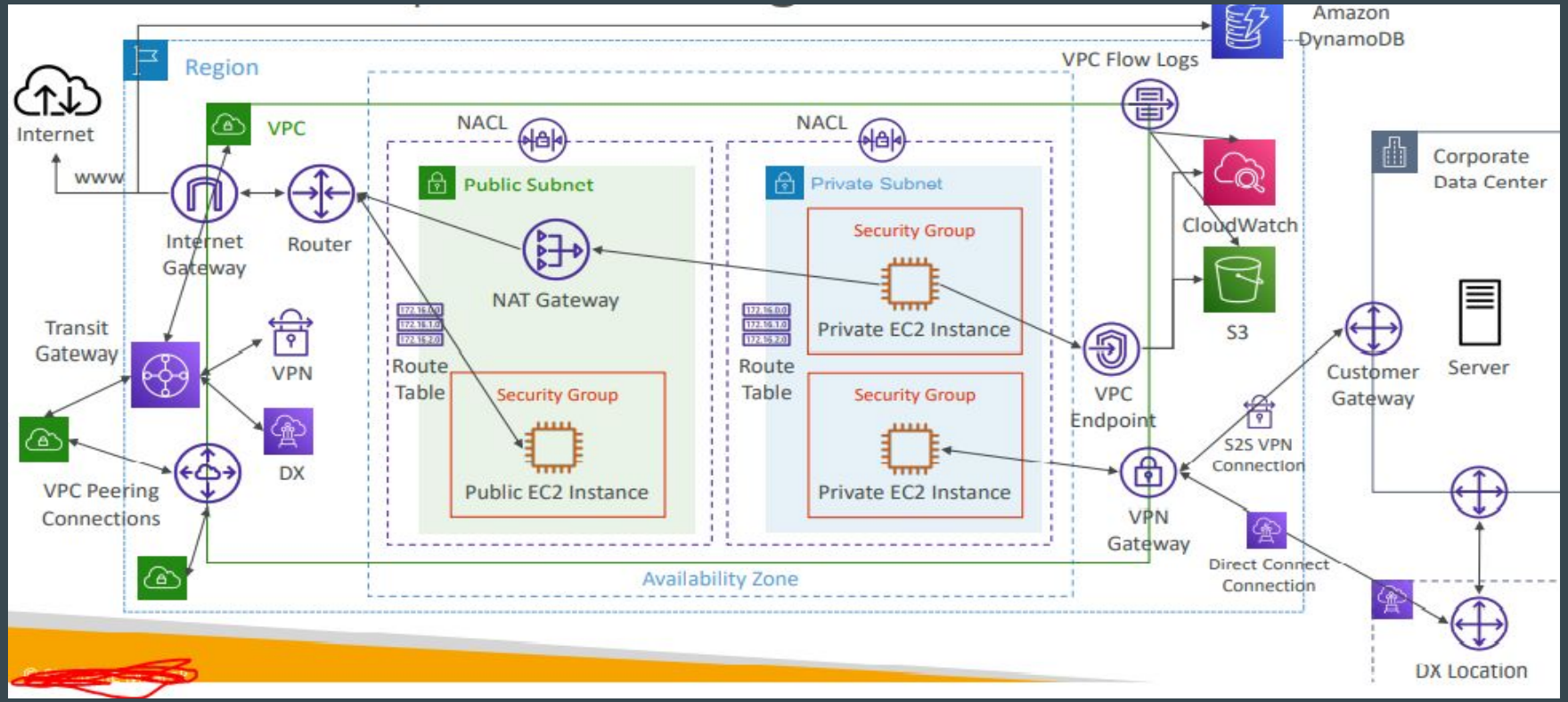
Virtual Private Cloud (VPC)

By Purusothaman S

VPC Components Diagram



Understanding CIDR – IPv4

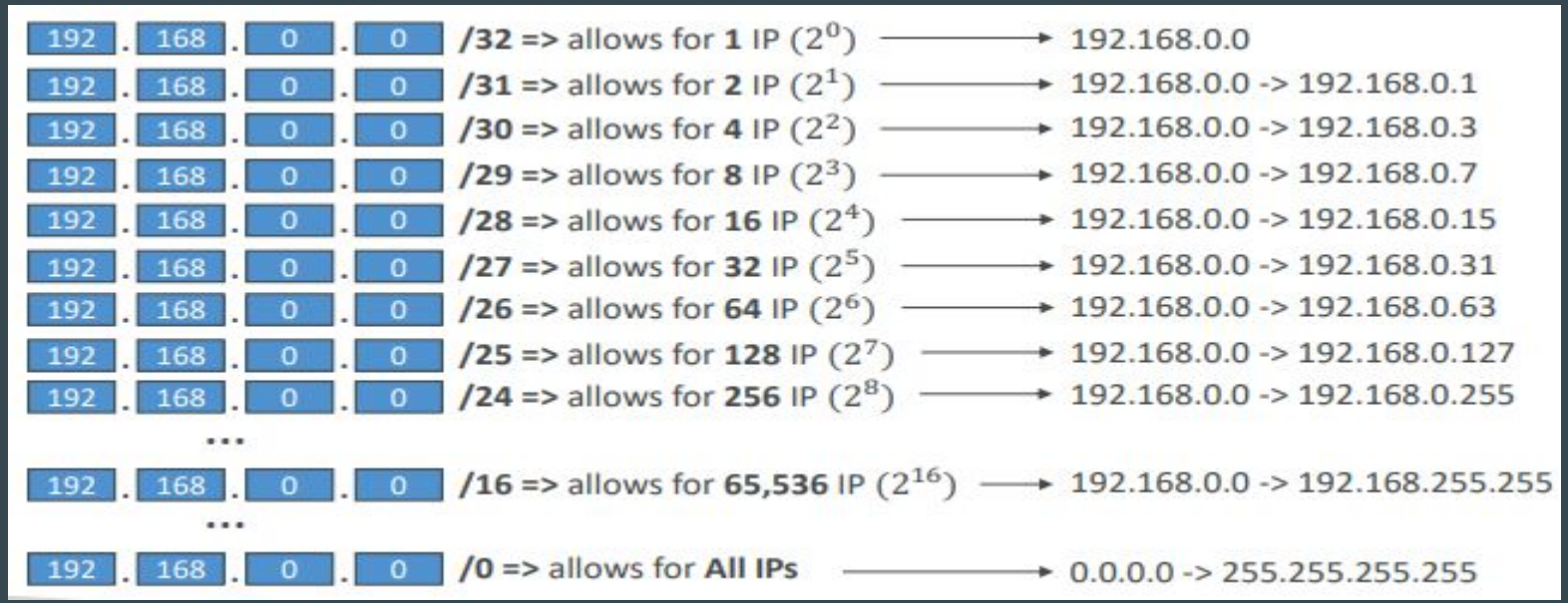
- Classless Inter-Domain Routing – a method for allocating IP addresses
- Used in Security Groups rules and AWS networking in general
- They help to define an IP address range:
- We’ve seen WW.XX.YY.ZZ/32 => one IP
- We’ve seen 0.0.0.0/0 => all IPs
- But we can define:192.168.0.0/26 =>192.168.0.0 – 192.168.0.63 (64 IP addresses)

Understanding CIDR – IPv4

- A CIDR consists of two components
- **Base IP** • Represents an IP contained in the range (XX.XX.XX.XX)
    - Example: 10.0.0.0, 192.168.0.0, ...
- **Subnet Mask**
    - Defines how many bits can change in the IP
        - Example: /0, /24, /32 • Can take two forms:
        - /8 255.0.0.0 • /16 255.255.0.0
        - /24 255.255.255.0 • /32 255.255.255.25

Understanding CIDR – Subnet Mask

The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP



Understanding CIDR – Little Exercise

- 192.168.0.0/24 = … ?
- 192.168.0.0/16 = … ?
- 134.56.78.123/32 = … ?
- 0.0.0.0/0
- All IPs! • When in doubt, use this website https://www.ipaddressguide.com/cidr

Public vs. Private IP (IPv4)

The Internet Assigned Numbers Authority (IANA) established certain blocks of IPv4 addresses for the use of private (LAN) and public (Internet) addresses

Private IP

- 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) ç in big networks
- 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) ç AWS default VPC in that range
- 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) ç e.g., home networks Consectetur ad

All the rest of the IP addresses on the Internet are Public

Default VPC Walkthrough

- All new AWS accounts have a default VPC
- New EC2 instances are launched into the default VPC if no subnet is specified
- Default VPC has Internet connectivity and all EC2 instances inside it have public IPv4 addresses
- We also get a public and a private IPv4 DNS name

VPC in AWS – IPv4

- **VPC = Virtual Private Cloud**
- You can have multiple VPCs in an AWS region (max. 5 per region – soft limit)
- Max. CIDR per VPC is 5, for each CIDR:
- **Min. size is /28 (16 IP addresses)**
- **Max. size is /16 (65536 IP addresses)**
- Because VPC is private, only the Private IPv4 ranges are allowed:

• 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) • 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) • 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)

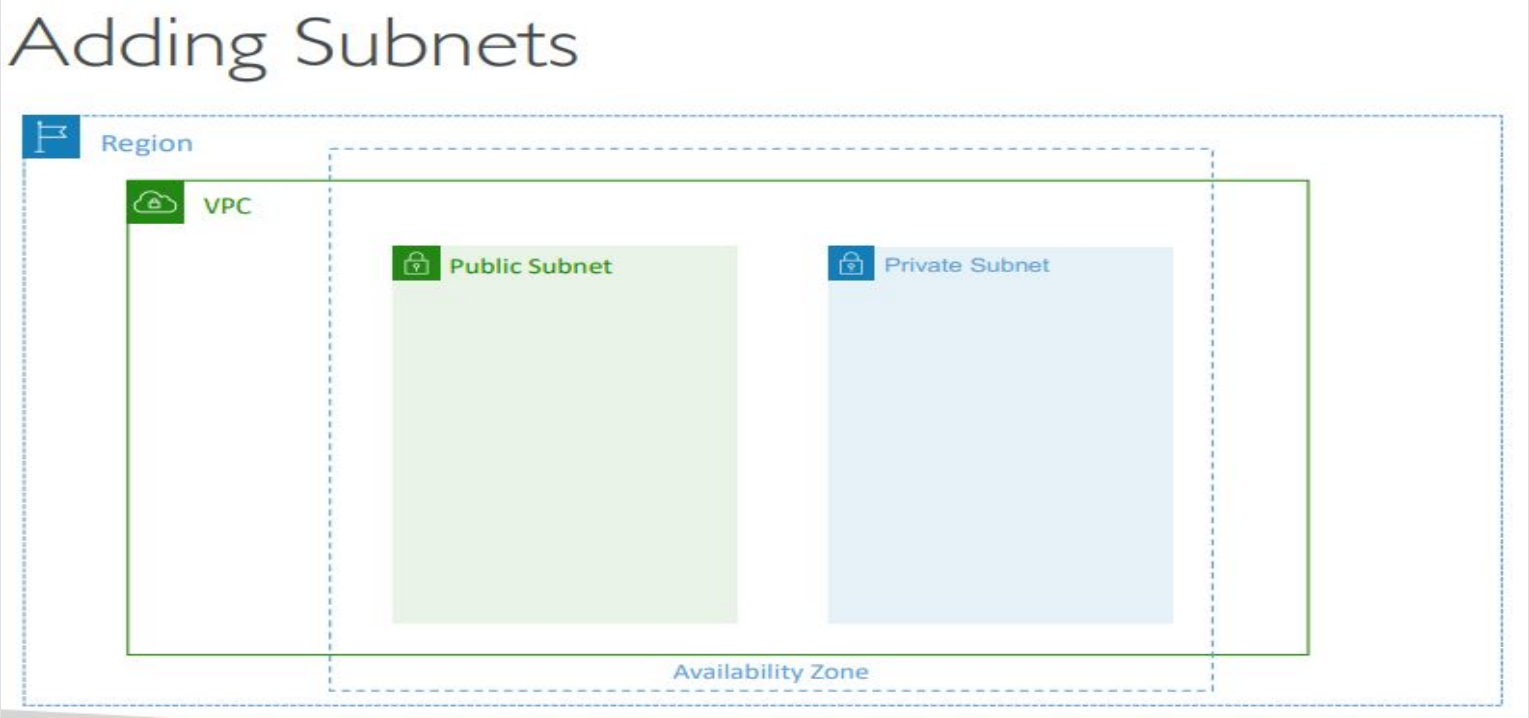
VPC – Subnet (IPv4)

- AWS reserves 5 IP addresses (first 4 & last 1) in each subnet

•• These 5 IP addresses are not available for use and can't be assigned to an EC2 instance

- Example: if CIDR block 10.0.0.0/24, then reserved IP addresses are:
  - 10.0.0.0 – Network Address
  - 10.0.0.1 – reserved by AWS for the VPC router
  - 10.0.0.2 – reserved by AWS for mapping to Amazon-provided DNS
  - 10.0.0.3 – reserved by AWS for future use
  - 10.0.0.255 – Network Broadcast Address. AWS does not support broadcast in a VPC

State of Hands -on



State of Hands -on

